

Homework 8

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Collaboration Statement

Problem 1 (*Cauchy Sequence: $s_n = \frac{1}{n}$*)

Prove, directly from the definitions, that (s_n) where $s_n = \frac{1}{n}$ is a Cauchy sequence.

Discussion.

So I initially wanted to prove that $s_n = \frac{1}{n}$ converges and then say by 10.9 Lemma that since it converges, it must be Cauchy. However, the problem specifically asks to prove it directly from the definitions, so I will need to work with the definition of a Cauchy sequence directly. So to review, here is the definition of a Cauchy sequences:

10.8 Definition.

A sequence (s_n) of real numbers is called a Cauchy sequence if for each $\varepsilon > 0$, there exists an N such that $m, n > N$ implies $|s_n - s_m| < \varepsilon$.

My plan is to show that for any given $\varepsilon > 0$, I can find an N such that for all $m, n \geq N$, the absolute difference $|s_n - s_m|$ is less than ε . I will manipulate the expression $|s_n - s_m| = \left| \frac{1}{n} - \frac{1}{m} \right|$ and find a suitable N based on ε that guarantees that $\frac{1}{n} + \frac{1}{m} < \varepsilon$ for all $m, n \geq N$.

To find such an N , notice that for $m, n \geq N$,

$$\left| \frac{1}{n} - \frac{1}{m} \right| = \left| \frac{1}{n} + \left(\frac{-1}{m} \right) \right| \leq \frac{1}{n} + \frac{1}{m}.$$

By triangle inequality.

Thus, if we can make $\frac{1}{m} + \frac{1}{n} < \varepsilon$, the Cauchy condition will hold. So the question becomes: how can we choose an N such that for all $m, n \geq N$, we guarantee $\frac{1}{m} + \frac{1}{n} < \varepsilon$? Well, we would just need to ensure that both $\frac{1}{m} < \frac{\varepsilon}{2}$ and $\frac{1}{n} < \frac{\varepsilon}{2}$. This would guarantee that their sum is less than ε .

Then we would just manipulate these inequalities to find a suitable N . More concretely though, for $\frac{1}{n} < \frac{\varepsilon}{2}$, how big would n need to be for $\frac{1}{n} < \frac{\varepsilon}{2}$ to hold? Rearranging gives $n > \frac{2}{\varepsilon}$. Similarly, for $\frac{1}{m} < \frac{\varepsilon}{2}$, we get $m > \frac{2}{\varepsilon}$. Thus, if we choose N to be any natural number greater than $\frac{2}{\varepsilon}$, then for all $m, n \geq N$, both inequalities will hold.

But then how could we know that we can find such an N ? We know we can find a natural number N such that $N > \frac{2}{\varepsilon}$. Well, by the Archimedean property, for any real number $\frac{2}{\varepsilon}$ there exists a natural number N such that

$$N > \frac{2}{\varepsilon}.$$

Then, for all $m, n \geq N$ we have $\frac{1}{m} \leq \frac{\varepsilon}{2}$ and $\frac{1}{n} \leq \frac{\varepsilon}{2}$, so that

$$\frac{1}{n} + \frac{1}{m} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, this choice of N guarantees that $|s_n - s_m| < \varepsilon$ for all $m, n \geq N$.

Proof.

Let $\varepsilon > 0$ and $N > \frac{2}{\varepsilon}$. Then, for all $m, n > N$,

$$|s_n - s_m| = \left| \frac{1}{n} - \frac{1}{m} \right| = \left| \frac{1}{n} + \left(\frac{-1}{m} \right) \right| \leq \frac{1}{n} + \frac{1}{m} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, (s_n) is a Cauchy sequence. □

Problem 2 (*Cauchy Sequence: $x_n = \sqrt{n}$*)

Let $x_n = \sqrt{n}$. Show that while the difference between consecutive terms goes to 0, i.e., show $\lim |x_{n+1} - x_n| = 0$, the sequence (x_n) is not a Cauchy sequence.

Discussion.

Proof.

□

Problem 3 (*Ross Chapter 2 10.6 (a)*)

Let (s_n) be a sequence such that $|s_{n+1} - s_n| < 2^{-n}$ for all $n \in \mathbb{N}$. Prove that (s_n) is a Cauchy sequence and hence a cin

Discussion.

Proof.

□

Problem 4 (*Ross Chapter 2 14.6*)

- (a) Prove that if $\sum |a_n|$ converges and (b_n) is a bounded sequence, then $\sum a_n b_n$ converges.
Hint: Use Theorem 14.4.
- (b) Observe that Corollary 14.7 is a special case of part (a).

Discussion.

Proof.

□

Problem 5 (*Ross Chapter 2 14.7*)

Prove that if $\sum a_n$ is a convergent series of nonnegative numbers and $p > 1$, then $\sum a_n^p$ converges.

Discussion.

Proof.

□

Problem 6 (*$\varepsilon - \delta$ Convergence Proof*)

Prove using the $\varepsilon - \delta$ definition of the limit of a function that if $\lim_{x \rightarrow a} f(x) = L$, then $\lim_{x \rightarrow a} 4f(x) = 4L$.

Discussion.

Proof.

□